

Customer Intelligence & Segmentation Analysis

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End-to-End Customer Analytics and Segmentation Project

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Tools & Technologies

• Python • Pandas • NumPy • Matplotlib • Seaborn • Scikit-Learn • Power BI • VS Code • Jupyter Notebooks

1. Project Background

Customer segmentation is one of the most important analytical techniques used by modern businesses to understand customer behavior and optimize marketing strategies.

Rather than treating every customer the same, organizations group customers based on purchasing patterns, spending habits, engagement levels, and overall business value. This enables targeted marketing campaigns, improved customer retention, and more efficient allocation of business resources.

The objective of this project was to build a complete customer intelligence solution that transforms raw transactional data into actionable business insights.

The project was developed using Python and Machine Learning in Visual Studio Code and later visualized through an interactive Power BI dashboard.

2. Project Objectives

- Analyze customer purchasing behavior.
- Identify high-value and low-value customer groups.
- Measure customer engagement using behavioral metrics.
- Create meaningful customer segments.
- Generate business recommendations for each segment.
- Build an executive-level Power BI dashboard for decision makers.

3. Dataset Overview

Total Customers Analyzed: 4,338

Features:

CustomerID, Recency, Frequency, Monetary Value, Product Diversity, Tenure, Average Order Value, Average Basket Size.

4. Notebook 01 – Data Cleaning & Preparation

Activities:

- Missing value assessment
- Duplicate record detection
- Data type validation
- Outlier inspection
- Data standardization

Outcome:

A clean and reliable dataset was prepared for downstream analytics and machine learning.

5. Notebook 02 – Exploratory Data Analysis (EDA)

Analyzed:

- Recency
- Frequency
- Monetary Value
- Tenure

Key findings:

- Most customers purchased infrequently.
- Revenue distribution was highly skewed.
- A small customer group generated significant revenue.

6. Notebook 03 – Feature Engineering

Created:

- Recency
- Frequency
- Monetary
- Average Order Value
- Average Basket Size
- Product Diversity
- Customer Tenure

Outcome:

Enhanced customer representation for segmentation.

7. Notebook 04 – Customer Segmentation Using Machine Learning

Features used:

- Recency
- Frequency
- Monetary Value
- Product Diversity
- Tenure
- Average Order Value
- Average Basket Size

Process:

- Feature Scaling
- K-Means Clustering
- Cluster Profiling
- Business Label Assignment

8. Customer Segments Identified

VIP Customers

Loyal Customers

Potential Loyalists

Dormant Customers

Wholesale Buyers

9. Power BI Dashboard Development

Page 1 – Executive Summary

Page 2 – Segment Analysis

Page 3 – Customer Behavior Analytics

Page 4 – Strategic Recommendations

10. Key Business Insights

- VIP Customers generated the highest value.
- Loyal Customers contributed the largest recurring revenue.

- Potential Loyalists represented the strongest growth opportunity.
- Dormant Customers posed the highest retention risk.
- Wholesale Buyers generated large revenues through fewer transactions.

11. Business Impact

- Improved customer retention strategy
- Increased customer lifetime value understanding
- Reduced churn risk identification
- Data-driven decision support

12. Conclusion

This project transformed raw customer transaction data into actionable business intelligence through data cleaning, exploratory analysis, feature engineering, machine learning segmentation, and Power BI dashboard development.